Department of Computer Science and Engineering

UIET, CSJM University, Kanpur

ISCS-101 – Mid Semester Examination

(CSE-1 & CSE-2)

Duration: 90 Minutes | Maximum Marks: 30 | Session: 2026-27

Branch / Section: _____	**Date:** _____

Instructions: Attempt all questions. Answer all subpart of a question at same place. Show essential reasoning.

Section A – Answer all questions ($9 \times 1 = 9$ marks)

1. Determine the output: `int a = 23, b = 5, c = 2; printf("%d", a / b * c + a % b);`

2. Fill in the blanks so that this program reads two integers and stores their fractional average: `int total, count; float average; scanf("%d %d", ___, ___); average = ___ total / count;`

3. Determine the output: `int total = 23, count = 4; float x = total / count + (float)total / count; printf("%.2f", x);`

4. Determine the output: `int x = 031, y = 0x19; printf("%d", x - y);`

5. Starting from directory `~/A/B/C`, execute `cd ../../; mkdir Test; cd Test/../B`. What path will `pwd` display?

6. Determine the output: `int x = 7, y = 3; x += y * 2 - x % y; y = x / (y + 1); printf("%d %d", x, y);`

7. Determine the output: `char ch = 'F'; int n = ch - 'A'; printf("%d", n * (n - 1));`

8. Fill in the blanks to print a floating-point average with exactly two digits after the decimal point: `float average = (float)marks / count; printf("Average = _____", _____);`

9. A record uses `char section; int roll; float marks; double fee;`. Ignoring alignment and padding, how many complete records can fit in 64 bytes?

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Section B – Answer all questions ($3 \times 3 = 9$ marks)

10. Starting in the home directory, trace the commands below. Draw the final directory tree of `~/MidSem`. Then write one relative-path command to move from `Backup` to `Source`.

```
mkdir MidSem
cd MidSem
mkdir Source Backup
cd Source
touch answer.c rough.c
cp answer.c ../Backup
mv rough.c practice.c
cd ../Backup
touch readme.txt
```

11. Fill in the three blanks in this program. Then state the output for input `23 4`. Briefly state why the cast is needed.

```
int total, count;
float average;

scanf("%d %d", &total, &count);
average = _____ total / count;
printf("Average = _____\n", _____);
```

12. Identify and correct **any three** independent errors in the following program. Write the corrected form of each line.

```
#include <stdio.h>

int main(void)
{
    int hours, minutes, total;
    scanf("%d %d", hours, &minutes);
    total == hours * 60 + minutes;
    printf("Total minutes = %f\n", total);
    return 0
}
```

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Section C – Answer all questions ($2 \times 6 = 12$ marks)

13. Write a complete C program that reads the marks (out of 100) obtained in Quiz, Lab, and End-Term components. Calculate and display the weighted score using 20% Quiz, 30% Lab, and 50% End-Term weightage. The result must retain its decimal part. Do not use loops or conditional statements. Use meaningful variable names and suitable format specifiers.

Your output must clearly display: the three input marks, the weighted contribution of each component, and the final weighted score. **(6 marks)**

14. Study the following code. Show the values after each assignment and answer all parts.

```
int p = 7, q = 3, r = 2;
double x = 1.5, y;

p = p * q - r;
q = p % 4 + q;
x = (double)p / q + r;
y = x * q - p;

printf("p = %d, q = %d, x = %.2f, y = %.2f", p, q, x, y);
```

(a) Determine the exact output and show the calculation leading to it. **(2 marks)**

(b) Present the values of `p`, `q`, `x`, and `y` in a table after their respective assignments. **(2 marks)**

(c) Explain the purpose of `(double)p` in the calculation of `x`. Also state the value of `x` if that cast is removed. **(1 mark)**

(d) If `q = p % 4 + q;` is replaced by `q = p / 4 + q;`, calculate the new value of `q` and the resulting value of `x`. Show the relevant calculation. **(1 mark)**

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